

DSAA3073: Theories in Data Science

Week 10 Validation Sheet · Online Convex Optimization, Stability, and the Batch Boundary

Coverage: Week 10

Duration: 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. In online convex optimization, which benchmark is valid for a realized loss sequence $f_1, \dots, f_T$ when attainment has not been established?

- (a) $\sum_t f_t(x_t) - \min_{x_t \in \mathcal{K}} \sum_t f_t(x_t)$, with a different comparator each round
- (b) $\sum_t f_t(x_t) - \inf_{x \in \mathcal{K}} \sum_t f_t(x)$, with one fixed x on the same sequence
- (c) Compare with losses chosen on a second, more favorable sequence
- (d) Reveal f_t , choose its minimizer, and then record that choice as x_t

A2 [5 marks]. Let $p = \Pi_{\mathcal{K}}(y)$ for a nonempty closed convex set. Which statement is licensed by the Week 10 projection result?

- (a) $\|p - u\|_2 \leq \|y - u\|_2$ for every feasible $u \in \mathcal{K}$
- (b) Projection is linear-time for every representation of $\mathcal{K}$
- (c) The same distance inequality holds for every $u \in \mathbb{R}^d$
- (d) A nearest point is automatically unique for every nonconvex set

A3 [5 marks]. Which update respects the online information order for the registered linearized FTRL method?

- (a) $x_t = \arg \min_x \left[\eta \sum_{s=1}^t g_s^T x + \mathcal{R}(x) \right]$
- (b) $x_t = \arg \min_x \left[\eta \sum_{s=1}^t g_s^T x + \mathcal{R}(x) \right]$
- (c) $x_t = \arg \min_x \left[\eta \sum_{s=1}^t f_{s(x)} + \mathcal{R}(x) \right]$, declared identical to the registered linearized form
- (d) $x_t = \arg \min_x f_{t(x)}$ before f_t is revealed

A4 [5 marks]. Which statement correctly reconciles the general square-root lower bound with faster rates?

- (a) The lower bound rules out logarithmic regret for every restricted loss class.
- (b) A fixed strongly convex regularizer alone gives logarithmic regret for arbitrary convex losses.
- (c) The lower bound concerns the unrestricted bounded convex-loss class; extra loss curvature or exp-concavity with a matching method may permit logarithmic regret.
- (d) If $D = 0$ or $G = 0$, one should still substitute $\eta = \frac{D}{G\sqrt{T}}$.

Part B: Definitions in use

B1 [20 marks]. Work on $\mathcal{K} = [-1, 1]$. The realized linear losses are $f_{s(x)} = g_s x$, where $g_1 = \frac{1}{2}$, $g_2 = -1$, and $g_3 = 1$. Let $\mathcal{R}(x) = \frac{x^2}{2}$, $\eta = \frac{1}{2}$, and let the learner use the registered FTRL rule.

Verify directly that $\mathcal{R}$ is 1-strongly convex; compute the round-4 decision; and identify the fixed comparator for the three realized losses, stating why a minimum rather than only an infimum is legal here. Finally, audit the proposed alternative $\tilde{x}_4 = \arg \min_{x \in \mathcal{K}} [\eta \sum_{s=1}^4 g_s x + \mathcal{R}(x)]$: state exactly which information boundary it violates.

Answer space

Part C: Basic skills

C1 [7 marks]. In $\mathcal{K} = [0, 1]^2$, let $x_t = (0.8, 0.2)$, $g_t = (1, -2)$, and $\eta = \frac{1}{2}$. Compute the raw OGD proposal and the projected next point. For $u = (0.3, 0.7)$, calculate both squared distances needed to check that the projection step has not increased distance to this feasible comparator.

Answer space

C2 [8 marks]. The constant-step OGD bound is

$$R_T \leq \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}.$$

For $D = 6$, $G = 3$, and $T = 400$, find the positive step size that balances the two terms, compute each term, and report the resulting regret bound and average-regret bound $\frac{R_T}{T}$.

Answer space

Part D: Construction and diagnosis

D1 [17 marks]. On $\mathcal{K} = [-2, 2]$, FTL uses a fixed tie rule $x_1 = 2$. Let $f_1(x) = \frac{x}{4}$, and for $t \geq 2$ let $f_{t(x)} = -\frac{x}{2}$ when t is even and $f_{t(x)} = \frac{x}{2}$ when t is odd.

Determine the FTL decisions through round 5, then give a general argument that the decisions alternate between the endpoints. Compute the learner's cumulative loss, the best fixed comparator loss, and the regret for both even and odd T . Explain why changing only the round-1 tie rule cannot repair the asymptotic instability.

Answer space

D2 [18 marks]. A report argues: “FTRL adds a fixed strongly convex regularizer, so for every sequence of arbitrary convex losses its regret is automatically $O(\log T)$.” The only applicable guarantee in the report is

$$R_{T(u)} \leq 2\eta G_{\mathcal{R}}^2 T + \frac{\Delta_{\mathcal{R}}}{\eta}, \quad \eta > 0,$$

where $\Delta_{\mathcal{R}} > 0$ and $G_{\mathcal{R}} > 0$ are constants independent of T .

Diagnose the claim by finding the best order in T licensed by this guarantee. Then use the Week 10 lower-bound boundary to explain why a fixed strong regularizer does not remove the general obstruction, and state one valid route to logarithmic regret with all essential conditions.

Answer space

Part E: Proof and derivation

E1 [25 marks]. Let $\mathcal{K} \subseteq \mathbb{R}^d$ be nonempty, closed, and convex with diameter $D > 0$. Convex differentiable losses satisfy $\|\nabla f_{t(x)}\|_2 \leq G$ on $\mathcal{K}$, where $G > 0$. A feasible map $P_t : \mathbb{R}^d \rightarrow \mathcal{K}$ obeys

$$\|P_{t(y)} - u\|_2^2 \leq \|y - u\|_2^2 + \xi_t$$

for every $y \in \mathbb{R}^d$ and $u \in \mathcal{K}$, where $\xi_t \geq 0$. The learner uses $x_{t+1} = P_{t(x_t - \eta g_t)}$ with constant $\eta > 0$ and $g_t = \nabla f_{t(x_t)}$. Before round 1, the learner knows a deterministic budget $|\Xi|_T$ satisfying $\sum_{t=1}^T \xi_t \leq |\Xi|_T$.

Develop one coherent audit: derive and legally tune a regret guarantee using only information available before round 1; identify the asymptotic threshold on $|\Xi|_T$ for the resulting average guarantee to vanish; then determine whether the failure at linear total error can be genuine. For the last task, choose and construct your own one-dimensional nonempty closed convex feasible set, initial point, bounded linear losses, globally defined feasible maps P_t , and errors ξ_t with total error $\Theta(T)$ but regret $\Theta(T)$. Verify the approximation rule for **every** ambient input y and feasible comparator u , prove the complete iterate sequence, identify the best fixed comparator, compute exact regret, and explain what the instance establishes about the threshold.

Answer space

Part F: Synthesis and limitation

F1 [15 marks]. A report says: “Any online method with sublinear regret automatically returns a last iterate that generalizes to a population, even for adaptively ordered data and a nonconvex hypothesis class. OGD, FTRL, linear MWU, and AdaBoost are therefore the same algorithm.”

Repair the population-risk claim by giving the Week 10 online-to-batch assumptions, returned hypothesis, and form of the high-probability bound. Then distinguish the four named methods by the object and update each uses; shared regret or potential ideas are not enough to declare identical iterates.

Answer space